Interacting Multipoles of Rank 1 and 2

$$+\frac{1}{3}\cdot T_{\alpha\beta\gamma}\mu^B_\alpha\Theta^A_{\beta\gamma} = R_z^{-7}\cdot\frac{1}{3}\cdot\left[\begin{array}{c} -15\cdot R_\alpha\cdot R_\beta\cdot R_\gamma\cdot\mu^B_\alpha\cdot\Theta^A_{\beta\gamma} \\ +3\cdot R_z^2\cdot R_\alpha\cdot\delta\left(\beta,\gamma\right)\cdot\mu^B_\alpha\cdot\Theta^A_{\beta\gamma} \\ +3\cdot R_z^2\cdot R_\beta\cdot\delta\left(\alpha,\gamma\right)\cdot\mu^B_\alpha\cdot\Theta^A_{\beta\gamma} \\ +3\cdot R_z^2\cdot R_\gamma\cdot\delta\left(\alpha,\beta\right)\cdot\mu^B_\alpha\cdot\Theta^A_{\beta\gamma} \end{array}\right] \tag{1}$$

Simplify deltas:

$$+\frac{1}{3}\cdot T_{\alpha\beta\gamma}\mu^B_\alpha\Theta^A_{\beta\gamma} = R_z^{-7}\cdot\frac{1}{3}\cdot\left[\begin{array}{c} -15\cdot R_\alpha\cdot R_\beta\cdot R_\gamma\cdot\mu^B_\alpha\cdot\Theta^A_{\beta\gamma} \\ +3\cdot R_z^2\cdot R_\alpha\cdot\delta\left(\beta,\beta\right)\cdot\mu^B_\alpha\cdot\Theta^A_{\beta\beta} \\ +3\cdot R_z^2\cdot R_\beta\cdot\delta\left(\alpha,\alpha\right)\cdot\mu^B_\alpha\cdot\Theta^A_{\alpha\beta} \\ +3\cdot R_z^2\cdot R_\gamma\cdot\delta\left(\alpha,\alpha\right)\cdot\mu^B_\alpha\cdot\Theta^A_{\alpha\gamma} \end{array}\right] \tag{2a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-7}\cdot\frac{1}{3}\cdot\left[\begin{array}{c} -15\cdot R_\alpha\cdot R_\beta\cdot R_\gamma\cdot\mu^B_\alpha\cdot\Theta^A_{\beta\gamma} \\ +3\cdot R_z^2\cdot R_\alpha\cdot\mu^B_\alpha\cdot\Theta^A_{\beta\beta} \\ +3\cdot R_z^2\cdot R_\beta\cdot\mu^B_\alpha\cdot\Theta^A_{\alpha\beta} \\ +3\cdot R_z^2\cdot R_\gamma\cdot\mu^B_\alpha\cdot\Theta^A_{\alpha\gamma} \end{array}\right] \tag{2b}$$

$$\xRightarrow{\text{reduced indices}} R_z^{-7}\cdot\frac{1}{3}\cdot\left[\begin{array}{c} -15\cdot R_\alpha\cdot R_\beta\cdot R_\gamma\cdot\mu^B_\alpha\cdot\Theta^A_{\beta\gamma} \\ +3\cdot R_z^2\cdot R_\alpha\cdot\mu^B_\alpha\cdot\Theta^A_{\beta\beta} \\ +3\cdot R_z^2\cdot R_\beta\cdot\mu^B_\alpha\cdot\Theta^A_{\alpha\beta} \\ +3\cdot R_z^2\cdot R_\beta\cdot\mu^B_\alpha\cdot\Theta^A_{\alpha\beta} \end{array}\right] \tag{2c}$$

$$\xRightarrow{\text{added up}} R_z^{-7}\cdot\frac{1}{3}\cdot\left[\begin{array}{c} -15\cdot R_\alpha\cdot R_\beta\cdot R_\gamma\cdot\mu^B_\alpha\cdot\Theta^A_{\beta\gamma} \\ +3\cdot R_z^2\cdot R_\alpha\cdot\mu^B_\alpha\cdot\Theta^A_{\beta\beta} \\ +6\cdot R_z^2\cdot R_\beta\cdot\mu^B_\alpha\cdot\Theta^A_{\alpha\beta} \end{array}\right] \tag{2d}$$

Simplify R:

$$+\frac{1}{3}\cdot T_{\alpha\beta\gamma}\mu^B_\alpha\Theta^A_{\beta\gamma} = R_z^{-7}\cdot\frac{1}{3}\cdot\left[\begin{array}{c} -15\cdot R_z\cdot R_z\cdot R_z\cdot\mu^B_z\cdot\Theta^A_{zz} \\ +3\cdot R_z^2\cdot R_z\cdot\mu^B_z\cdot\Theta^A_{\beta\beta} \\ +6\cdot R_z^2\cdot R_z\cdot\mu^B_\alpha\cdot\Theta^A_{z\alpha} \end{array}\right] \tag{3a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-7}\cdot\frac{1}{3}\cdot\left[\begin{array}{c} +6\cdot R_z^3\cdot\mu^B_\alpha\cdot\Theta^A_{z\alpha} \end{array}\right] \tag{3b}$$

Simplify Dipoles:

$$+\frac{1}{3}\cdot T_{\alpha\beta\gamma}\mu^B_\alpha\Theta^A_{\beta\gamma} = R_z^{-7}\cdot\frac{1}{3}\cdot\left[\begin{array}{c} +6\cdot R_z^3\cdot\mu^B_y\cdot\Theta^A_{yz} \end{array}\right] \tag{4a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-7}\cdot\frac{1}{3}\cdot[+0] \tag{4b}$$

Combining everything:

$$+\frac{1}{3}\cdot T_{\alpha\beta\gamma}\mu^B_\alpha\Theta^A_{\beta\gamma} = +0 \tag{5}$$